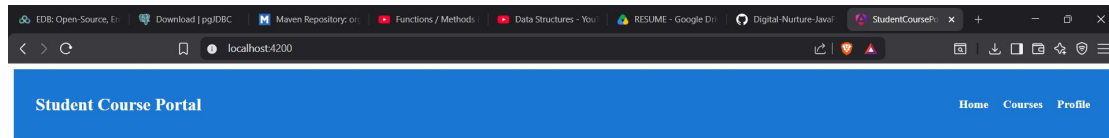


Angular

HANDS-ON 1 [Beginner]

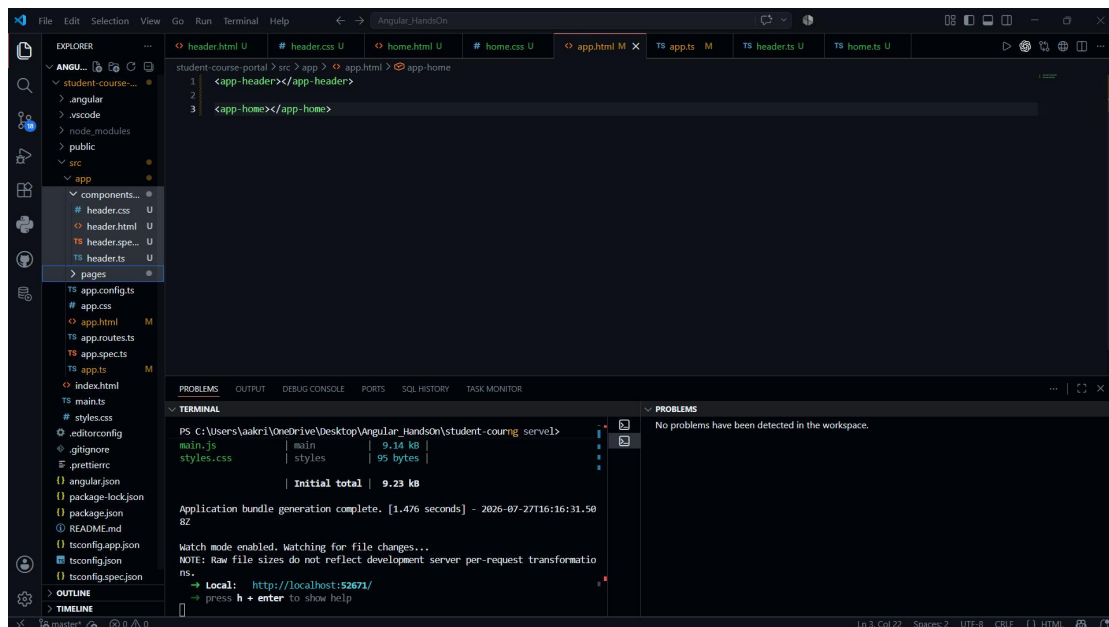
Environment Setup, Project Structure & First Component



Welcome to Student Course Portal

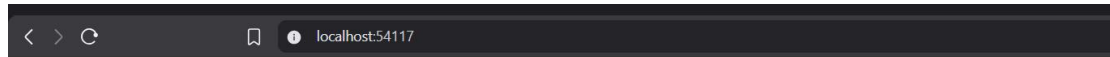
Manage your courses, profile and enrollments easily.

12 Courses Available	3 Enrolled	3.8 GPA
-------------------------	---------------	------------



HANDS-ON 2 [Beginner]

Data Binding, Lifecycle Hooks & Component Communication



Angular Hands-On 2

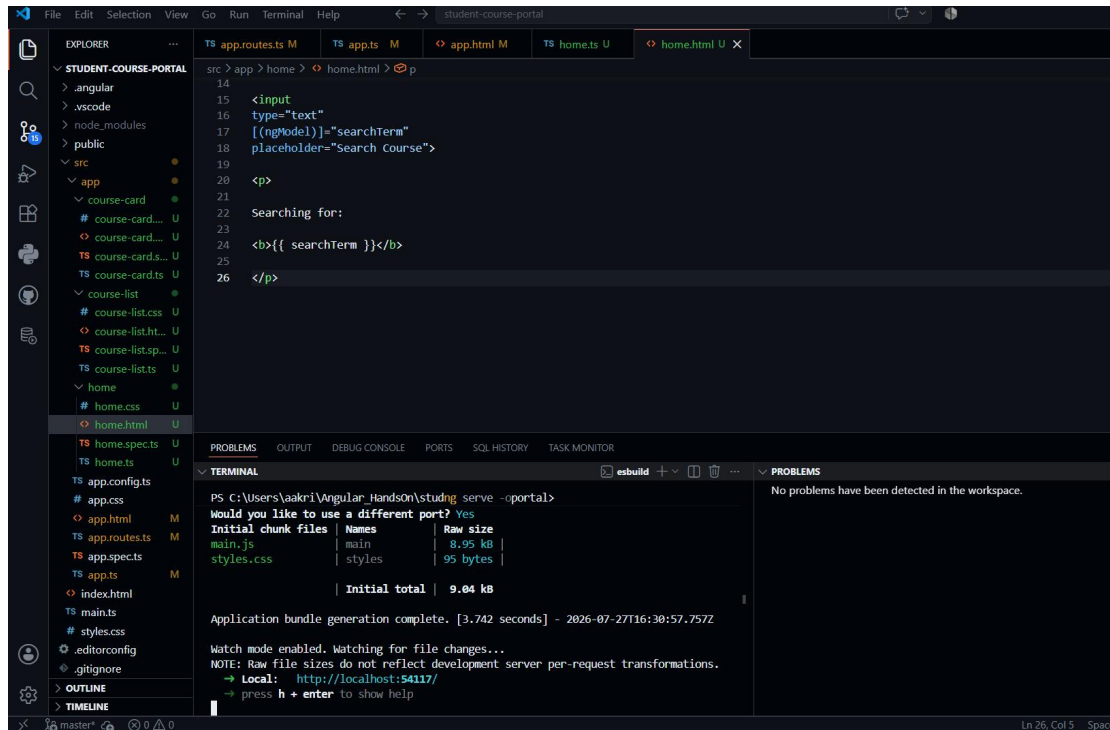
[Home](#) | [Courses](#)

Student Course Portal

Enroll Now

Search Course

Searching for:



HANDS-ON 3 [Beginner]

Directives & Pipes — Built-in and Custom

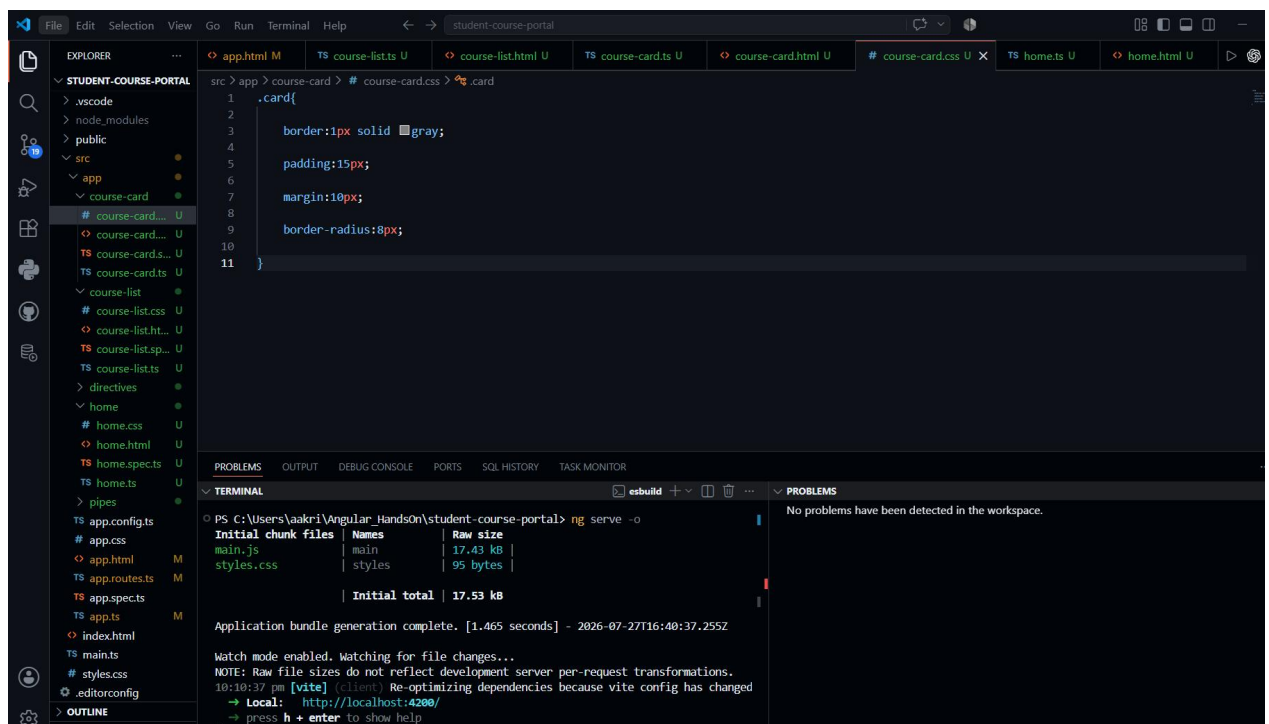
Angular Hands-On 2

[Home](#) | [Courses](#)

Available Courses

Loading courses...

No courses available.



HANDS-ON 4 [Intermediate]

Template-Driven Forms & Validation

EDB: Open-S... Download | p Maven Repo... Functions / h Data Structu... RESUME - G... Digital-Narr... StudentCour... StudentCour... StudentCour... Student... x +

localhost:4200/enroll

Student Course Portal

[Home](#) | [Courses](#) | [Enroll](#)

Enrollment Form

Name

Email

Course ID

Semester

☒ I Agree

The screenshot shows the VS Code editor with the following files open: enrollment-form.html, enrollment-form.css, course-lists, course-list.html, course-cards, course-card.html, and course-card.css. The enrollment-form.html file is selected, showing the following code:

```
1 <div>
2   <input type="text" value="AAKRISHYA"/>
3 </div>
4 <div>
5   <input type="text" value="aakrishya321mishra@gmail.com"/>
6 </div>
7 <div>
8   <input type="text" value="29939"/>
9 </div>
10 <div>
11   <input type="text" value="Odd"/>
12 </div>
13 <div>
14   <input checked="" type="checkbox"/> I Agree
15 </div>
16 <div>
17   <button type="button" value="Submit"/>
18   <button type="button" value="Reset"/>
19 </div>
20 </div>
```

The terminal output shows the following commands and results:

```
PS C:\Users\lakri\Angular_HandsOn\student-course-portal> ng g c enrollment-form
Initial chunk files | Names | Raw size
main.js | main | 31.19 kB
styles.css | styles | 95 bytes
Initial total | 31.28 kB

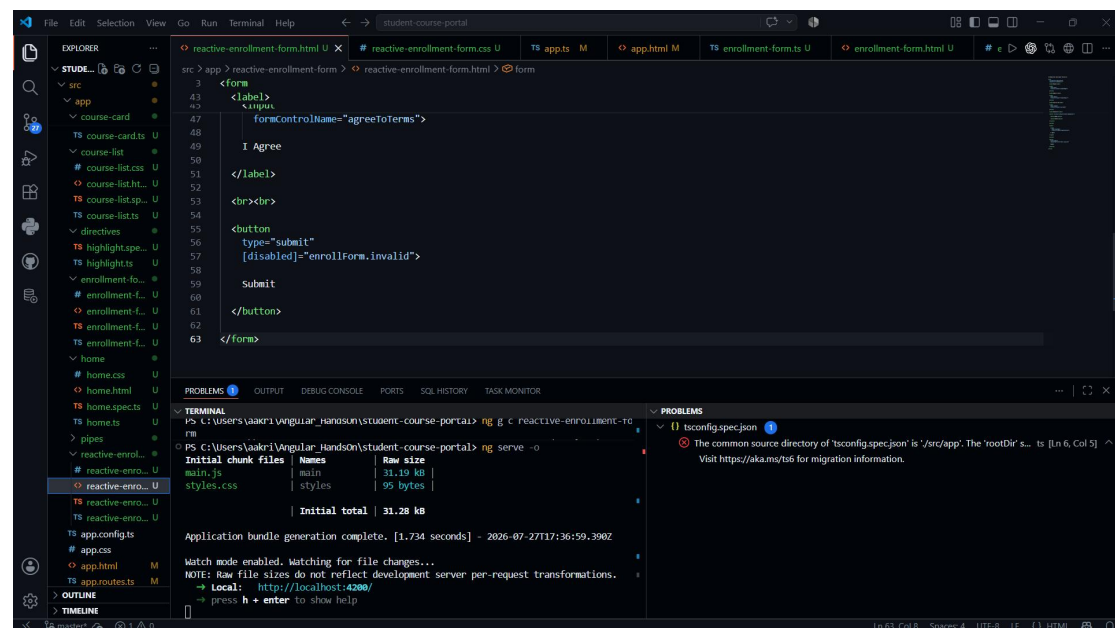
Application bundle generation complete. [1.700 seconds] - 2026-07-27T17:27:38.551Z

Watch mode enabled. Watching for file changes...
NOTE: Raw file sizes do not reflect development server per-request transformations.
-> Local: http://localhost:4200/
-> press h + enter to show help
```

The PROBLEMS panel shows no problems have been detected in the workspace.

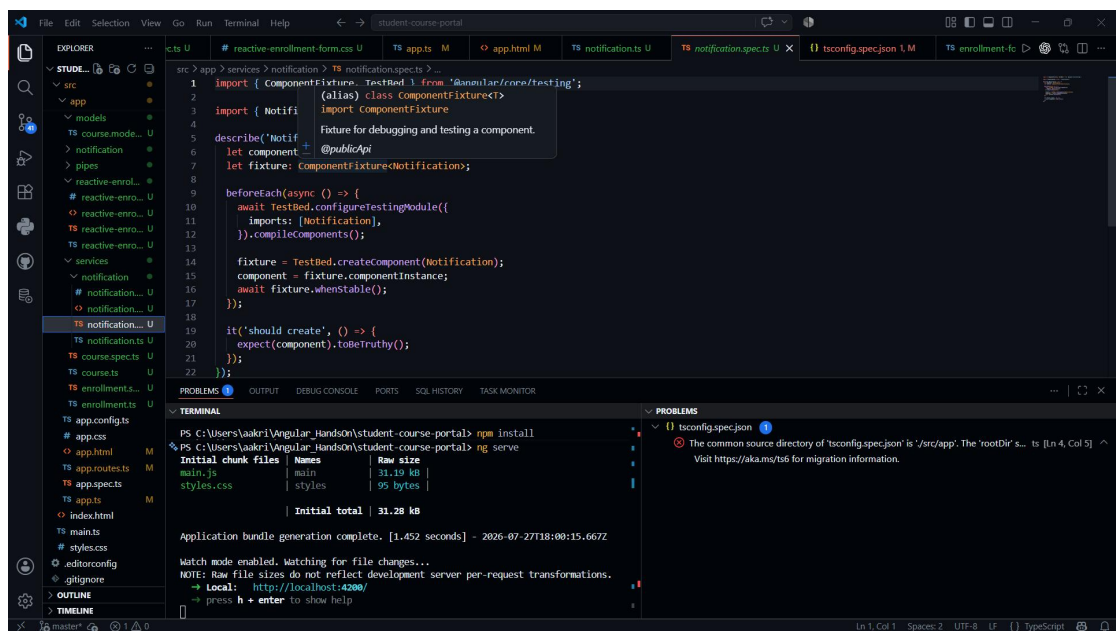
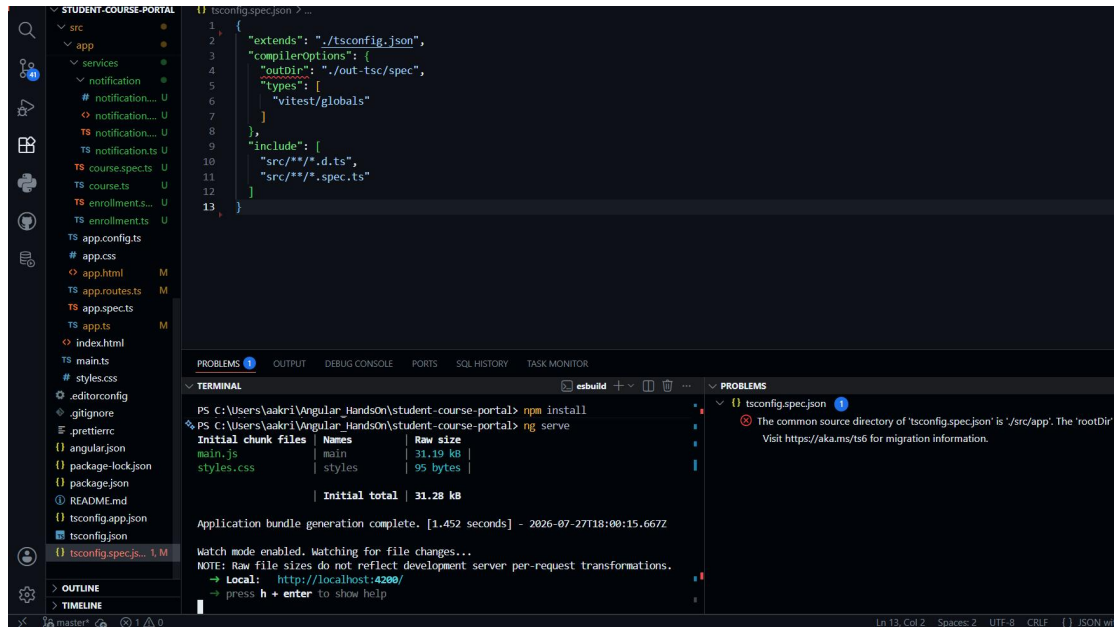
Reactive Forms — FormBuilder, FormGroup, FormArray & Custom

Enrollment request submitted successfully!



HANDS-ON 6 [Intermediate]

Services & Dependency Injection



HANDS-ON 7 [Intermediate]

Angular Routing — Guards, Lazy Loading & Route Data

The image shows a web browser at localhost:4200 displaying a "Student Course Portal". The page has a navigation bar with links for Home, Courses, and Profile. Below the navigation bar, there is an "Enroll Now" button. A message "Enrollment opened!" is displayed. A search bar contains the text "phy", and a message "Searching for: phy" is shown below it.

The VS Code editor shows the source code of the application. The Explorer panel on the left displays the project structure, including the src directory with files like app.html, app.ts, and app.spec.ts. The main editor shows the app.ts file, which defines the App component. The component has a selector of 'app-root', is standalone, and imports RouterOutlet and RouterLink. The templateUrl is './app.html' and the styleUrls are ['./app.css']. The export class App is defined.

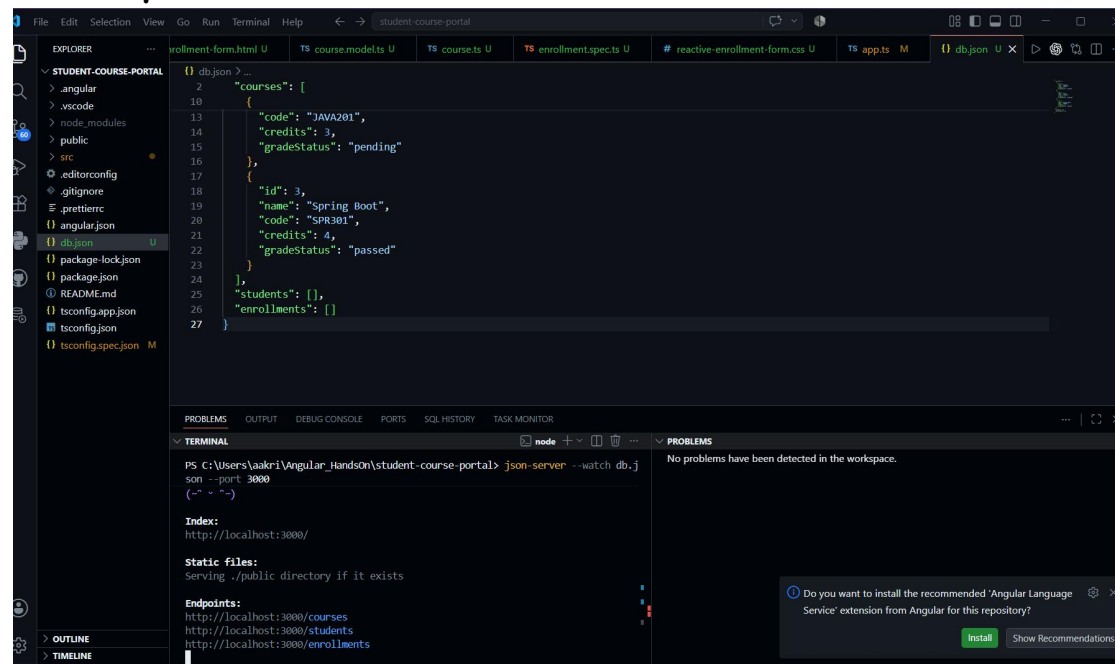
The TERMINAL panel at the bottom shows the output of the esbuild command. It displays the initial chunk files and their raw sizes:

Initial chunk files	Names	Raw size
main.js	main	23.14 kB
styles.css	styles	95 bytes
Initial total		23.23 kB

The terminal also shows the application bundle generation complete message, the watch mode enabled message, and the local URL: http://localhost:4200/. It also includes a note about raw file sizes and a prompt to press h + enter to show help.

HANDS-ON 8 [Advanced]

HTTP Client — API Integration, Observables & Interceptors



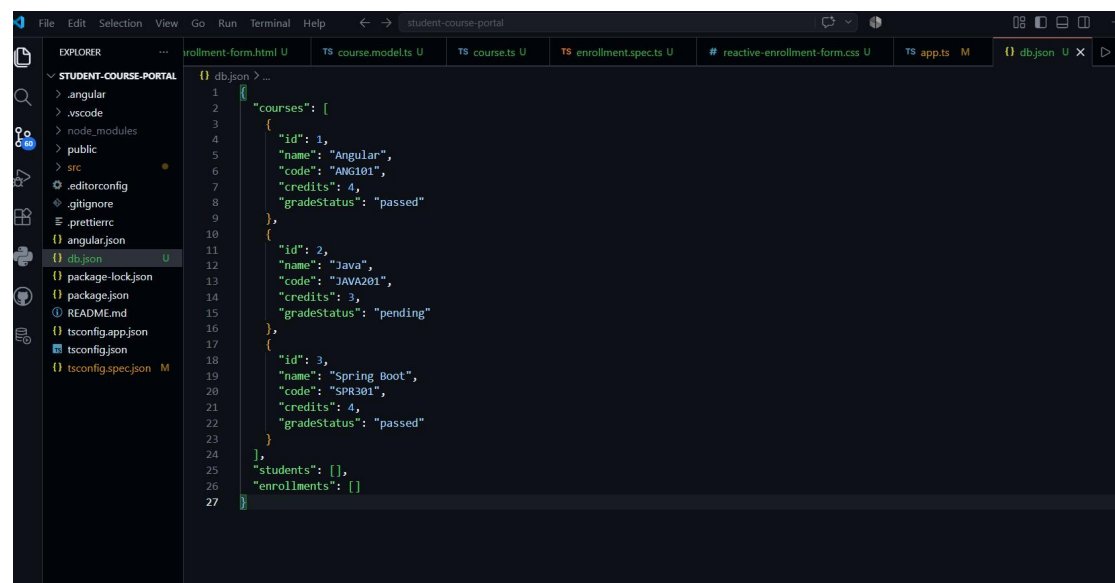
The screenshot shows the Visual Studio Code interface. The Explorer sidebar on the left displays the file structure of a project named 'STUDENT-COURSE-PORTAL'. The main editor area shows a file named 'db.json' with the following JSON content:

```
1 {  
2   "courses": [  
3     {  
4       "code": "JAVA201",  
5       "credits": 3,  
6       "gradeStatus": "pending"  
7     },  
8     {  
9       "id": 3,  
10      "name": "Spring Boot",  
11      "code": "SPR301",  
12      "credits": 4,  
13      "gradeStatus": "passed"  
14     }  
15   ],  
16   "students": [],  
17   "enrollments": []  
18 }  
19  
20  
21  
22  
23  
24  
25  
26  
27
```

The bottom panel shows the TERMINAL with the following output:

```
PS C:\Users\aaakri\Angular_HandsOn\student-course-portal> json-server --port 3000 db.json  
(~ ^ ~ ^)  
  
Index:  
http://localhost:3000/  
  
Static files:  
Serving ./public directory if it exists  
  
Endpoints:  
http://localhost:3000/courses  
http://localhost:3000/students  
http://localhost:3000/enrollments
```

A notification bubble in the bottom right corner asks: "Do you want to install the recommended 'Angular Language Service' extension from Angular for this repository?" with buttons for "Install" and "Show Recommendations".



The screenshot shows the Visual Studio Code interface with the same project structure. The main editor area shows a file named 'db.json' with the following JSON content:

```
1 {  
2   "courses": [  
3     {  
4       "id": 1,  
5       "name": "Angular",  
6       "code": "ANG101",  
7       "credits": 4,  
8       "gradeStatus": "passed"  
9     },  
10    {  
11      "id": 2,  
12      "name": "Java",  
13      "code": "JAVA201",  
14      "credits": 3,  
15      "gradeStatus": "pending"  
16    },  
17    {  
18      "id": 3,  
19      "name": "Spring Boot",  
20      "code": "SPR301",  
21      "credits": 4,  
22      "gradeStatus": "passed"  
23    }  
24  ],  
25  "students": [],  
26  "enrollments": []  
27 }
```

HANDS-ON 9 [Advanced]

State Management — NgRx Store, Actions, Reducers, Effects & Selectors

